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Subject: Application for the Internship "Population balance modelling of liquid-liquid extraction" (R12-4) — Direction Conception Modélisation Procédés, IFPEN Lyon

Dear Hiring Team,

I am a final-year Industrial Chemical Engineering student at INSAT (Université de Carthage, Tunisia), specializing in Process and Environment, and I am writing to apply for the internship "Population balance modelling of liquid-liquid extraction" (ref. R12-4) within the Direction Conception Modélisation Procédés at IFPEN Lyon. I am available from 1 February 2027, for 6 months (3 to 9 months possible), within your January–April 2027 window.

My coursework in thermodynamics, fluid mechanics and unit operations led me to an academic project on the prediction of liquid-liquid equilibria and cross-flow and countercurrent extraction for ternary systems: I formulated the LLE thermodynamic system, solved it with Rachford-Rice and arc-length continuation, reproduced 7 Mathematica notebooks proving the equivalence of the Hunter-Nash and McCabe-Thiele methods, and reimplemented the workflow in Python (SciPy), MATLAB and Mathematica. During my internship at SEGULA Technologies, I solved the PDEs of a Li-ion cell model by the finite volume method and validated it against experimental data, reducing the RMSE from 202.13 mV to 40.5 mV. Building a Python model and validating it against experimental data is exactly the workflow your internship calls for.

IFPEN's choice to predict droplet size distributions with a 0D/1D population balance, testing literature kernels against in-house ones, addresses a real design question for extraction columns, since drop size governs mass transfer.

I am drawn to this internship because it extends the extraction modelling I started at INSAT from equilibrium stages to droplet-scale physics.

I would welcome the opportunity to discuss my application further and am available for an interview at your convenience. Thank you for your consideration.

Sincerely,
Nermine Moumen